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EXERCISES

# Lecture 2: Recap

AI508 — Algorithmic Game Theory

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# 1 Probability

Serves to understand outcomes with some probability

## Definition: Sample space

The space which an event can hold.

## Example: Different sample spaces

- Coins  $\Omega = \{H, T\}$
- Die  $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Real values  $\Omega = \mathbb{R}$

## Definition: Event

An Event is a subset of the sampling space.

## Example: Different events

Given that  $A$  is an event in  $\Omega$  such that  $0 \leq P(A) \leq 1$ , then  $P(\Omega) = 1$  and  $P(\phi) = 0$

Remember that

$$P(A^C) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

## 1.1 Random variables

### Definition: Random Variable

The random variable  $X$  is a function that maps

$$\mathcal{X} : \Omega \rightarrow \mathbb{R}$$

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \cdot P(\mathcal{X} = x)$$

### Example: Coin

Two coins giving heads

$$\mathcal{X} = \{0, 1, 2\}$$

### Example: Discrete random variable

Given that  $\Omega = [0, 1]$ , then

$$X(\omega) = \begin{cases} -1 & w \leq \frac{1}{2} \\ 1 & w > \frac{1}{2} \end{cases}$$

### Definition: Probability mass function

Given  $\mathcal{X}$

$$P(\mathcal{X} = x) = p_x$$

### Definition: Joined distribution

Given  $\mathcal{X}, \mathcal{Y}$

$$P(\mathcal{X} = x, \mathcal{Y} = y) = p_{xy}$$

### Definition: Marginalization

Given  $\mathcal{X}, \mathcal{Y}$

$$P(\mathcal{X} = x) = \sum_{y \in \mathcal{Y}} P(\mathcal{X} = x, \mathcal{Y} = y)$$

### Definition: Bernoulli

Given that  $\mathcal{X} = \begin{Bmatrix} 1 & p \\ 0 & 1-p \end{Bmatrix}$

So that  $\mathbb{E}[\mathcal{X}] = 1p + 0(1-p) = p$

### Definition: Variance

Variance captures the spread of value around the

$$\text{Var}[\mathcal{X}] = \mathbb{E}[(\mathcal{X} - \mathbb{E}[\mathcal{X}])^2]$$

### Example: Variance on bernoulli

Given  $X$ ,

$$\text{Var}[X] = \sum_{x \in \mathcal{X}} (x - \mu)^2 P(X = x)$$

### Definition: Covariance

Given  $\mathcal{X}, \mathcal{Y}$ ,

$$\begin{aligned} \text{Cov}[\mathcal{X}, \mathcal{Y}] &= \mathbb{E}[(\mathcal{X} - \mathbb{E}[\mathcal{X}])(\mathcal{Y} - \mathbb{E}[\mathcal{Y}])] \\ &= \mathbb{E}[\mathcal{X}\mathcal{Y} + \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}] - \mathcal{X}\mathbb{E}[\mathcal{Y}] - \mathcal{Y}\mathbb{E}[\mathcal{X}]] \\ &= \mathbb{E}[\mathcal{X}\mathcal{Y}] + \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}] - \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}] - \mathbb{E}[\mathcal{Y}]\mathbb{E}[\mathcal{X}] \\ &= \mathbb{E}[\mathcal{X}\mathcal{Y}] - \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}] = \text{Cov}[\mathcal{X}] \end{aligned}$$

## 1.2 Conditional probability

Assume an implicit sample space.

### Definition: Conditional probability

Assume there are 2 events  $A$  and  $B$  and let  $P(B) > 0$

Then the probability of event  $A$  given  $B$  is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

### Example: Conditional probability

Given  $B = \{x > 3\} = \{4, 5, 6\}$

The probability that  $A = \text{EVEN} = \{2, 4, 6\}$  is:

$$\frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{6}}{\frac{3}{6}} = \frac{2}{3}$$

### Definition: Bayes Rule

Given  $A, B$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Where  $P(B|A)$  is the *likelihood*,  $P(A)$  is the *prior* and  $P(B)$  is the *marginal evidence*

### Definition: Independent events of a random variable

Event  $A$  and  $B$  are independent if and only if

$$P(A \cap B) = P(A)P(B)$$

### Example: Given that $B > 0$

Given that  $B > 0$  then

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

### Definition: Independence of 2 random variables

2 R.V's  $X, Y$  are independent iff

$$P(\mathcal{X} = x, \mathcal{Y} = y) = P(\mathcal{X} = x)P(\mathcal{Y} = y) \quad \forall x, y \in (x \in \mathcal{X}, y \in \mathcal{Y})$$

### Definition: Uncorrelation of 2 random variables

2 R.V's are uncorrelated iff

$$\text{Cov}(\mathcal{X}, \mathcal{Y}) = 0 \implies \mathbb{E}[\mathcal{X}\mathcal{Y}] = \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}]$$

### Example: Independence and Uncorrelation

Note that

$$\text{Independence} \implies \text{Uncorrelation}$$

## 2 Conditional expectation

Note that conditional expectations are not scalars,

### Definition: Conditional expectation

Given  $\mathcal{Y}$

$$\mathbb{E}[\mathcal{X}|\mathcal{Y} = y] = \sum_{x \in \mathcal{X}} x \cdot P(\mathcal{X} = x|\mathcal{Y} = y)$$

### Theorem: Law of Total Expectation

Given  $X$  and  $Y$ ,

$$\begin{aligned} \mathbb{E}[\mathcal{X}] &= \mathbb{E}[\mathbb{E}[\mathcal{X}|\mathcal{Y} = y]] \\ &= \sum_{y \in \mathcal{Y}} \left( \sum_{x \in \mathcal{X}} x \cdot P(\mathcal{X} = x|\mathcal{Y} = y) \right) P(\mathcal{Y} = y) \end{aligned}$$

### 3 Inequalities

#### Definition: Markov's inequality

Let  $\mathcal{X}$  be a non negative random variable, then.

$$P(X \leq a) \leq \frac{\mathbb{E}[\mathcal{X}]}{a}$$

#### Proof of Markov's inequality